

JAVA PROGRAMMING LABORATORY

Experiment No: 05

Date: 21/09/26

Title: Demonstration of SortedSet Methods Using TreeSet in Java

1. Objective / Aim

To write and execute a Java program that demonstrates the use of the **SortedSet** interface with **TreeSet**, and to understand commonly used methods such as `add()`, `first()`, `last()`, `headSet()`, `tailSet()`, `subSet()`, and `comparator()`.

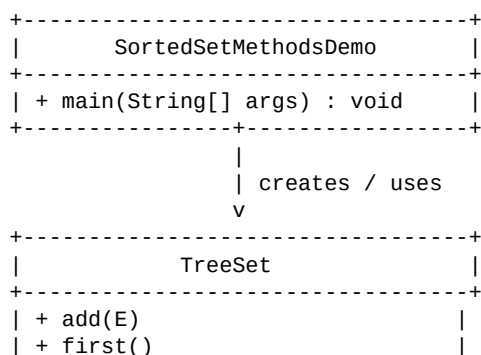
2. System Requirements

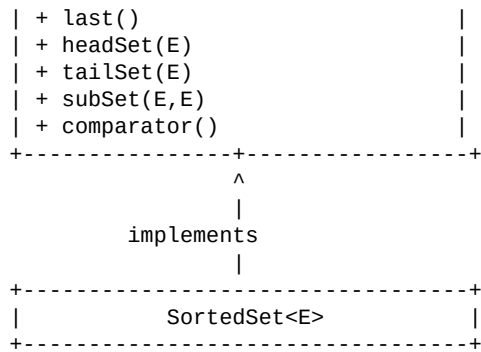
Hardware Requirements	Software Requirements
Computer/Laptop	Operating System: Windows / Linux / macOS
Processor: Dual Core or above	JDK: 17 or above (JDK 21 recommended)
RAM: 4 GB or above	Java compiler (javac) and Java Runtime Environment (java)
Storage: At least 1 GB free space	IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code (optional)

3. Algorithm / Logic Description

1. Start the program.
2. Import the SortedSet interface and TreeSet class.
3. Create a SortedSet of Integer values using TreeSet.
4. Add the values 40, 10, 70, 20, 50, and 30 to the set.
5. Display the complete SortedSet. TreeSet automatically stores the values in ascending order.
6. Use `first()` to obtain the smallest element.
7. Use `last()` to obtain the largest element.
8. Use `headSet(40)` to obtain elements strictly less than 40.
9. Use `tailSet(30)` to obtain elements greater than or equal to 30.
10. Use `subSet(20, 60)` to obtain elements from 20 (inclusive) to 60 (exclusive).
11. Use `comparator()` to display the comparator used for ordering. The natural ordering returns null.
12. Display all results and stop the program.

4. UML Class Diagram





5. Source Code

```

import java.util.SortedSet;
import java.util.TreeSet;

class SortedSetMethodsDemo {

    public static void main(String[] args) {

        // Creating a SortedSet using TreeSet
        SortedSet<Integer> numbers = new TreeSet<>();

        // Adding elements
        numbers.add(40);
        numbers.add(10);
        numbers.add(70);
        numbers.add(20);
        numbers.add(50);
        numbers.add(30);

        System.out.println("SortedSet: " + numbers);

        // first()
        System.out.println("First element: " + numbers.first());

        // last()
        System.out.println("Last element: " + numbers.last());

        // headSet()
        System.out.println("HeadSet before 40: " + numbers.headSet(40));

        // tailSet()
        System.out.println("TailSet from 30: " + numbers.tailSet(30));

        // subSet()
        System.out.println("SubSet from 20 to 60: " + numbers.subSet(20, 60));

        // comparator()
        System.out.println("Comparator: " + numbers.comparator());

    }
}

```

6. Completion and Execution Commands

Step 1: Save the program with the file name:
SortedSetMethodsDemo.java

Step 2: Open Command Prompt / Terminal in the folder containing the file.

Step 3: Compile the program:
javac SortedSetMethodsDemo.java

Step 4: Execute the program:
java SortedSetMethodsDemo

7. Output

SortedSet: [10, 20, 30, 40, 50, 70]

```
First element: 10
Last element: 70
HeadSet before 40: [10, 20, 30]
TailSet from 30: [30, 40, 50, 70]
SubSet from 20 to 60: [20, 30, 40, 50]
Comparator: null
```

8. Result / Completion

The Java program was successfully written, compiled, and executed. The program demonstrates the SortedSet interface using TreeSet and shows sorted storage along with first(), last(), headSet(), tailSet(), subSet(), and comparator() methods.